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RESEARCH ARTICLE

Loss of forest cover leads to population isolation in an imperiled, headwater-specialist fish

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Abstract

Context Anthropogenic land use degrades matrix, habitat, and corridors and can disrupt dispersal rates or dispersal attempts, resulting in population connectivity patterns that reflect the cost (or resistance to dispersal) imposed from degraded conditions (isolation by resistance; IBR) rather than that of geographic distance alone (isolation by distance; IBD).

Objectives As a small-bodied, benthic riverine species with relatively low dispersal capability, the imperiled Kentucky Arrow Darter (KAD), *Etheostoma spilotum*, is vulnerable to the negative effects of anthropogenic land use on population connectivity. To inform conservation efforts for KAD, we identified the primary drivers of population isolation (IBD vs. IBR) in the species using a riverscape genetics framework.

Methods We created resistance surfaces from landscape variables that may influence KAD occurrence and dispersal success (elevation, presence of dams, stream order, and percent landcover type). Cumulative resistance was correlated with F_{ST} values using a linear mixed effects model in R package corMLPE.

Results Though we found significant, positive relationships between genetic and geographic distances, analyses of single and multi-surface resistance models demonstrated that isolation by resistance was the primary driver of genetic distances among KAD populations. In particular, decreased forest cover was linked to high genetic differentiation.

Conclusions These results illustrate that population connectivity primarily degrades through resistance to dispersal rather than geographic distance between populations. Conservation efforts should focus on mitigating past and ongoing land use associated with deforestation, in addition to genetic rescue, as habitat restoration will not likely occur quick enough to prevent continued declines in KAD populations.

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Keywords Riverscape genetics · Gene flow · Dispersal · Population connectivity · Metapopulation dynamics · Isolation by Resistance

Introduction

Gene flow, the movement of genes from one population to another, is a crucial evolutionary force promoting genetic homogenization and limiting genetic differentiation among populations. Lack of gene flow results in population isolation, leading to loss of genetic diversity from processes like genetic drift and inbreeding. Inbreeding depression and declines in genetic diversity can lower the adaptive potential of a species and limit its ability to survive stochastic environmental changes, ultimately increasing risk of extinction (Keller and Waller 2002; Frankham 2005; Nabutanyi and Wittmann 2021). The negative effects associated with isolation from lack of gene flow are more pronounced on smaller populations, making imperiled species with declines in population size more at risk. Since the negative effects associated with gene flow are more pronounced on small populations and result in increased extinction risk, effective conservation management of imperiled species should prioritize understanding and promoting gene flow among populations.

The main mechanism of gene flow among populations is successful dispersal from one population to another. Successful dispersal is dependent on individual dispersal ability (i.e., determined by physiological constraints), matrix and habitat characteristics (e.g., frequency, quality, and distance between habitat patches and corridors), and landscape features (e.g., topography, hydrology, and geology; [Ebenhard 1991; McCullough and Wildlife Society 1996; Vandermeer and Carvajal 2001]). Anthropogenic land uses such as urbanization, mineral exploitation, and deforestation further reduce or eliminate dispersal in impacted areas. Such land uses often degrade or eliminate habitat, dispersal corridors, and matrix quality, leading to increased habitat fragmentation, declines in species occurrence and abundance, and reduced recolonization potential for extirpated populations (Dodd 1990; Harrison and Bruna 1999; Brauer and Beheregaray 2020); the resulting increased population isolation from lack of gene flow can lead to loss of genetic diversity through genetic drift and inbreeding (Fahrig and Merriam 1985; Stacey and Taper 1992; Harrison and Taylor 1997; Méndez et al. 2014; Fuller et al. 2015).

Biological or life history traits also influence a species' response to habitat fragmentation (Ewers and Didham 2007; Farneda et al. 2015; Keinath et al. 2017). Particularly, riverine-adapted organisms have naturally restricted dispersal pathways and may be more negatively impacted by corridor degradation and habitat fragmentation than terrestrial taxa (Fagan 2002; Fausch et al. 2002; Fuller et al. 2015; Grummer et al. 2019; Brauer and Beheregaray 2020). For example, benthic, headwater-stream adapted fishes with low dispersal potential, such as many darter species (Percidae: Etheostomatinae), are more sensitive to habitat fragmentation and have an increased risk of extirpation and extinction relative to larger-bodied pelagic fishes with higher dispersal potential (Benz and Collins 1997; Fagan 2002; Gallagher et al. 2015; Bland 2017).

The Kentucky Arrow Darter (KAD), *Etheostoma spilotum,* (Gilbert 1887; Bailey 1948; Page and Burr 2011; Thomas et al. 2023) is a small-bodied, benthic headwater-specialist that lacks a swimbladder and has a relatively short larval drift phase (Wallus and Simon 2006), making its survival dependent on clean substrates for foraging, spawning and shelter (Thomas et al. 2023). These benthic adaptations, along with its small body size, reduce its overall dispersal potential and ability to bypass instream features (Camak and Piller 2018), making the species vulnerable to population isolation from anthropogenic habitat and landscape alterations. Historically, KAD populations were widely distributed across the upper Kentucky River system in Kentucky (Thomas 2008) and maintained gene flow across their range (Blanton et al. 2019). Surveys have shown declines in occurrence and abundance, with documented extirpations from 36 of 74 historically occupied streams ([USFWS] United States Fish and Wildlife Service 2016; Floyd 2022). These declines led to the listing of KAD as federally threatened under the Endangered Species Act in 2016 ([USFWS] United States Fish and Wildlife Service 2016). An evaluation of contemporary levels of genetic diversity and population structure in KAD using 11 microsatellite loci revealed a high degree of structure among and low genetic diversity within populations (Blanton et al. 2019). The authors inferred human-mediated habitat degradation had

likely led to increased isolation of the populations and negative impacts to genetic diversity.

The landscape spanning the historic distribution of KAD in the upper Kentucky River system is naturally forested and mountainous (Figure S1). Before European settlement, forest cover spanned over 90,000 km^2 across Kentucky, but by the early 2000's approximately 50,000km^2 remained (Li et al. 2022). Surface coal mining, urbanization, agriculture, and river impoundments have negatively impacted streams on the Cumberland Plateau in the native range of KAD in eastern Kentucky through deforestations and modifications to instream chemical and physical properties, such as elevated specific conductance (conductivity) and sedimentation levels (Hitt et al. 2016, 2020; Cianciolo et al. 2020; Figure S1). These land use alterations have created poor habitat and corridor conditions for many stream reaches and have contributed to declines in KAD occurrence and abundance, perhaps due to genetic isolation (Hitt et al. 2016, 2020; Blanton et al. 2019; Floyd 2022; Fig. 1). A riverscape genetics study examining the effects of landscape features (slope, elevation and stream size) and anthropogenically altered habitat features (conductivity, culverts and forest cover) on population connectivity of KAD found that high conductivity levels from coal mining in the region best explained population structure at a small-spatial scale (i.e. within a single stream system; Watson et al. 2024). However, the influence of both natural and human-mediated landscape features on population connectivity and genetic diversity of KAD range-wide has not been studied.

Small-bodied, benthic, riverine species with relatively low dispersal capability, such as the KAD, are expected to exhibit isolation by distance (IBD) and follow a stepping-stone gene flow model, where genetic exchange among populations decreases with increasing geographic distance (Kimura and Weiss 1964; Slatkin 1993). However, habitat fragmentation in human-altered landscapes can disrupt these expected patterns

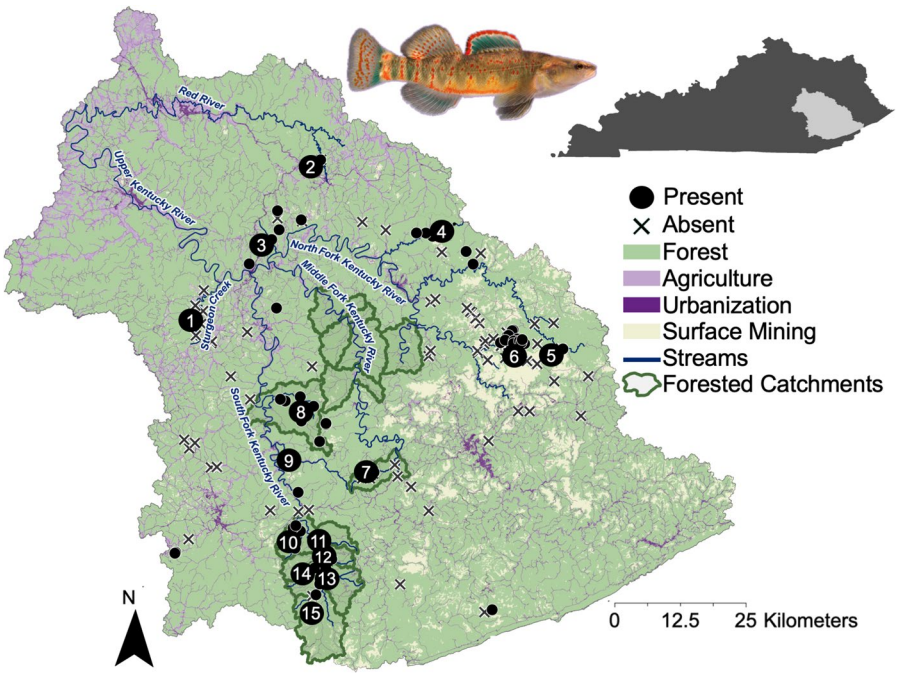

Fig. 1 Distribution of extant (black circles, 'present') and extirpated (x, 'absent) populations of Kentucky Arrow Darter (*Etheostoma spilotum*) and anthropogenic land use variables. Numbered, black circles represent sample collection sites for riverscape genetic models. HUC12 catchments outlined in green are the only catchments with forest cover ninety percent or greater. Sites with numbers are those sampled for genetic data used in our riverscape genetics modeling of *E. spilotum*. Numbers denote: 1.Wild Dog Creek, 2. Rockbridge Fork, 3. Hell Creek, 4. Frozen Creek, 5. Middle Fork Quicksand Creek, 6. Clemons Fork, 7. Hell For Certain Creek, 8. Laurel Fork, 9. Bullskin Creek, 10. Big Double Creek, 11. Gilbert's Big Creek, 12. Elisha Creek, 13. Bowen Creek, 14. Spring Creek, 15. Upper Bear Creek

by altering dispersal rates or dispersal attempt success (Coulon et al. 2010; Fletcher et al. 2018). As a result, gene flow patterns may reflect the cost (or resistance to dispersal) imposed from degraded habitat, matrix, or corridors, rather than geographic distance alone, leading to genetic differentiation from isolation by resistance (IBR). Similarly, isolation by barrier (IBB) assumes that a physical barrier will block dispersal and ultimately increase genetic differentiation between populations (Manel et al. 2003; Euclide and Marsden 2018). Naturally occurring landscape features such as a large waterfall may lead to IBB, but artificial impoundments are the predominant physical barrier to aquatic species' dispersal and a leading cause of habitat fragmentation in rivers (Warren et al. 2000; Dudgeon et al. 2006; Barletta et al. 2010; Reis et al. 2016). Therefore, freshwater fishes like the KAD may show genetic differentiation patterns that reflect IBR or IBB instead of or in addition to the expected IBD.

To identify the primary drivers of population isolation in KAD and better inform conservation for the species, we applied genetic analyses and riverscape resistance modeling to assess the relative contributions of IBD, IBR, and IBB on population connectivity. Given that Blanton et al. (2019) found higher levels of genetic differentiation than expected under a strict IBD model in KAD populations, we hypothesized that KAD dispersal and therefore connectivity are influenced by anthropogenic land use. We tested the potential effects of natural and anthropogenic landscape variables on population connectivity, including sites separated by long distances with low expectations for connectivity and proximate populations with expected connectivity. We predicted that KAD genetic differentiation will reflect IBR and/or IBB, where population connectivity is decreased by historic and current anthropogenic-mediated landscape conditions. Specifically, we expected river impoundments and land uses that contribute to decreased forest cover have degraded matrix, corridor, and habitat conditions, resulting in population isolation from decreased dispersal and reduced gene flow.

Methods

Site selection and tissue collection

We collected tissue samples from 15 sites in the upper Kentucky River system that best represent the extant distribution of the species (Fig. 1). Collections of KAD individuals occurred from November 2020 to November 2021 using standard backpack electrofishing methods. For all captured individuals, a non-lethal fin clip was taken from the caudal fin and stored in 95% ethanol for DNA preservation. Tissues were collected under appropriate collection permits and Austin Peay State University IACUC #20–015.

Microsatellite genotyping

We extracted whole genomic DNA from fin clips using the Qiagen DNeasy Blood and Tissue Kit (QIAGEN, Inc.) following manufacturer's instructions and diluted DNA to 10 ng μl^{-1}. We amplified 11 microsatellite loci (Blanton et al. 2019) in three multiplex reactions. Polymerase chain reaction (PCR) was performed in 10uL reactions, consisting of 5.65 µl of molecular-grade water, 1.00 µl 10X standard Taq reaction buffer (no MG), 1.20 µl MgCl2, 0.20 µl dNTPs, 0.25 µl forward primer, 0.50 µl reverse primer, 0.10 Taq Polymerase (New England Biolabs, Inc.), 0.10 µl M13 ABI dye-labeled primer, and 1.00 µl of 10 ng μl^{-1} DNA stock. PCR cycle conditions started with a denaturation step of 94 °C for 1 min, followed by 35 cycles of 94 °C for 30 s, 60 °C (or 57 °C for primer D116) for 30 s, and 72 °C for 30 s, with a final extension at 72 °C for 5 min. A negative control (no DNA) and three positive controls were included on each plate. Gel electrophoresis of all PCR product was used to ensure successful amplification, and the negative control was included to confirm there was no contamination in PCR reactions. We analyzed PCR products on an ABI 3730 sequencer with LIZ600 size standard at the DNA Core Sequencing Facility at the University of Illinois, Urbana-Champagne. Positive control samples were used to maintain consistency in scoring of allele calls across runs and scored genotypes using Genemarker v. 2.7 (SoftGenetics, LLC.). Our genetic data from South Fork Kentucky sites included genotypes from Watson et al. 2024, which were rescored to ensure consistency in allele calls (Table 1).

Table 1 Genetic diversity metrics for populations used to model the effects of landscape variables on genetic differentiation (F_{ST}) of Kentucky Arrow Darter (*Etheostoma spilotum*). Site number (#) and name correspond to sites sampled and are used to identify populations. Drainage indicates the watershed for each population examined (ST = Sturgeon Creek, RR = Red River, NF = North Fork Kentucky River, MF = Middle Fork Kentucky River, SF = South Fork Kentucky River). Diversity metrics examined using microsatellite data include number of individuals genotyped (N), rarified allelic richness (A_R), mean number of alleles per locus (N_a), observed (H_o) and expected (H_e) heterozygosity, p-values from Hardy–Weinberg equilibrium (HWE) exact tests (significant values in bold; * denotes significance after Bonferroni Correction), inbreeding coefficient (F_{IS}; significant values in bold), and inbreeding coefficient lower bound (F_{IS} L-CI) and upper bound (F_{IS} U-CI) confidence intervals

Site #	Population Name	Drainage	N	A_R	N_a	H_o	H_e	HWE	F_{IS}	F_{IS} L-CI	F_{IS} U-CI
1	Wild Dog Creek	ST	18	5.2	5.4	0.65	0.64	0.37	0.02	-0.08	0.05
2	Rockbridge Fork	RR	20	1.4	1.5	0.08	0.07	1.00	-0.02	-0.23	0.18
3	Hell Creek	NF	19	3.6	3.8	0.52	0.51	0.23	0.01	-0.11	0.07
4	Frozen Creek	NF	21	3.5	3.7	0.49	0.47	0.33	-0.02	-0.14	0.06
5	Middle Fork Quicksand Creek	NF	23	6.2	7.0	0.66	0.64	0.33	-0.01	-0.09	0.03
6	Clemons Fork	NF	21	3.3	3.5	0.35	0.37	0.45	0.07	-0.05	0.15
7	Hell For Certain Creek	MF	27	2.5	2.7	0.26	0.30	**< 0.01***	**0.17**	**0.00**	**0.29**
8	Laurel Fork	SF	20	4.5	4.7	0.59	0.55	0.89	**-0.05**	**-0.16**	**0.00**
9	Bullskin Creek	SF	16	3.6	3.6	0.51	0.46	0.09	-0.06	-0.20	0.00
10	Big Double Creek	SF	19	4.2	4.4	0.46	0.54	**< 0.01***	**0.17**	**0.05**	**0.23**
11	Gilbert's Big Creek	SF	19	3.8	3.9	0.51	0.50	0.99	0.02	-0.10	0.07
12	Elisha Creek	SF	15	3.7	3.7	0.42	0.41	0.88	-0.01	-0.16	0.05
13	Bowen Creek	SF	20	3.4	3.5	0.56	0.52	0.48	-0.05	-0.17	0.01
14	Spring Creek	SF	20	3.7	3.9	0.43	0.42	0.42	0.01	-0.12	0.09
15	Upper Bear Creek	SF	17	3.8	3.9	0.43	0.49	**< 0.01***	0.15	-0.02	0.25

Marker validation, population structure, and genetic diversity

We tested microsatellite data for evidence of null alleles, large allele drop-out, and stutter using MICROCHECKER v. 2.2.3 with 1,000 simulations and 95% confidence intervals to examine significance. Deviation from Hardy–Weinberg Equilibrium was tested for all sites and loci using the function "test_hw" and linkage disequilibrium was assessed to confirm loci independence using function "test_LD" in *GenePop* v. 4.7.5. Both linkage disequilibrium and deviation from Hardy–Weinberg equilibrium were tested with default Markov Chain parameters (dememorization = 1000, batches = 100, batch iterations = 1000) and type I error was reduced by Bonferroni correction using the "p.adjust" function in *stats* (Cabin and Mitchell 2000, R 2022).

We assessed genetic diversity for all sites using a suite of standard population-genetic metrics in the R package *diversity* (Keenan et al. 2013; Table S1). We used the function "divBasic" to generate mean number of alleles (N_a), observed heterozygosity (H_o), expected heterozygosity (H_e), allelic richness (A_R), and inbreeding coefficient (F_{IS}). We used rarefication when calculating A_R to account for uneven sample sizes at each site. The "popgenreport" function in the R package *PopGenReport* (Adamack and Gruber 2014) was used to evaluate private alleles and their frequencies.

We evaluated genetic differentiation and isolation between each site by calculating pairwise F_{ST} values with methods described by Weir and Cockerham (1984) using the function "pairwise.wcfst" in the R package *hierfstat* (Goudet 2005; Table S1). We calculated confidence intervals using the function "boot.ppfst" with 10,000 bootstraps in *hierfstat* and calculated p-values in GenoDive to evaluate deviation from panmixia.

Isolation by distance

We tested for IBD in KAD to determine if there is a statistically significant relationship between genetic differentiation (F_{ST}) and geographic distance among populations. We calculated river distances between

sites in the R package *riverdist* v 0.15.5 (Tyers 2017); river distance (km) was log transformed to identify potential outliers and reduce point spread. We performed an isolation by distance Mantel test in R using the package *adegenet* v 2.1.8 (Jombart 2008) with a reduced major axis regression to assess the strength of the relationship between genetic differentiation and river distance (Jensen et al. 2005).

Landscape variable selection and resistance classification

To identify potential landscape drivers of genetic differentiation (pairwise F_{ST}) and putative population isolation in KAD, we selected seven variables known from existing literature to influence fish dispersal or KAD occurrence, including Strahler stream order, slope, river impoundments, and four categories of land cover (forest, urbanization, agriculture, and surface mining). For each of our seven landscape variables, except river impoundments (see below), we generated rasters from the National Landcover Dataset (NLCD; Dewitz and Geological Survey 2024) and from Pericak et al. (2018) in ArcMap 10.6.1 (ESRI 2011) to test their potential effects on observed genetic differentiation in KAD. Using information from previous studies (Table S2), we reclassified each landscape variable raster to represent resistance values, where higher values represent landscape characteristics that restrict movement, and lower values represent landscape characteristics that facilitate movement. Resistance values ranged from 1 to 100, where the lowest resistance (cell value = 1) represented the facilitation of movement, and the highest resistance (cell value = 100) represented the total impediment of movement (SI Appendix A). All rasters were cropped to include only the study area, represented by a 150 m buffer of National Hydrography Dataset Plus (NHDPlus HR; U.S. Geological Survey 2019) stream networks in the existing range of KAD in ArcMap 10.6.1 (Fig. 2).

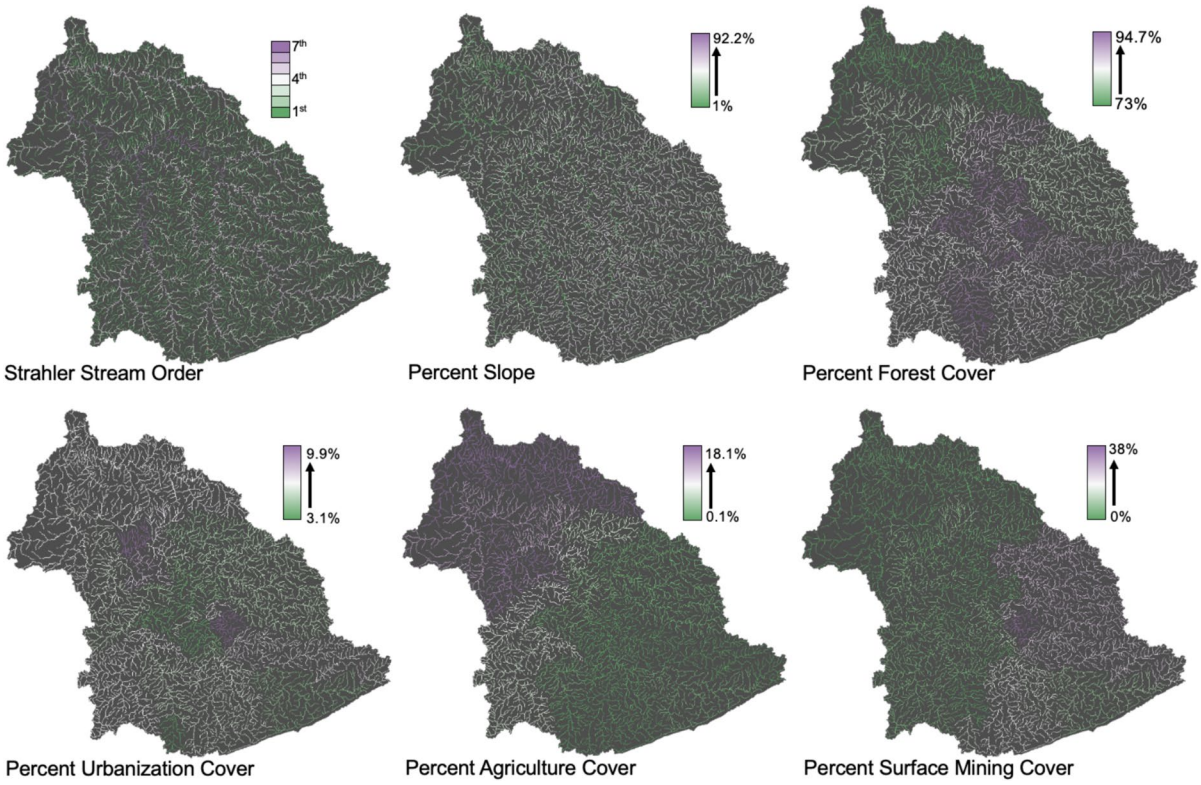

Fig. 2 Landscape variable raster layers used in riverscape genetics modeling of Kentucky Arrow Darter (*Etheostoma spilotum*)

River impoundments

River impoundments destruct natural hydrological and ecological processes at a large spatial scale (Baxter 1977; Graf 1999; Mbaka and Wanjiru Mwaniki 2015; Winton et al. 2019) and lead to genetic discontinuities among populations that were once historically connected prior to impoundment construction (Wu et al. 2003; Fluker et al. 2014; Sotola et al. 2017; Barnett et al. 2020). To account for the potential effects of river impoundments on population connectivity of KAD, we used a presence/absence modeling approach since only one river impoundment was present in the study system on the Middle Fork Kentucky River. All pairwise comparisons with populations that would have to travel through a river impoundment were assigned a value of 1, and all pairwise comparisons that would not have to travel through a river impoundment were assigned a value of 0. (Fig. 2)

Riverscape analyses

We used a riverscape resistance modeling approach to test for IBR and examine the relative effects of landscape variables on genetic differentiation of KAD. Resistance modeling uses pairwise resistance distance (i.e., the sum of resistance raster cells across all possible paths between two sites, where higher values represent movement restriction and lower values represent movement facilitation) to examine relative ease of dispersal between populations (McRae 2006). Since landscape variables often have combined effects on the environment, we stacked all possible combinations of our landscape variable resistance rasters to examine the potential combined effects of our landscape variables on KAD gene flow and dispersal. We used the "commuteDistance" function in the R package *gdistance* (Van Etten 2017) to calculate pairwise resistance distance between all populations across all landscape variables both independently and as stacked combinations. As this function takes conductance values (1/resistance) rather than resistance values as input, we calculated the inverse of each cell value of all resistance layers.

We used generalized linear mixed-effect models with a generalized least squares (GLS) and maximum likelihood population effects (MLPE) approach in the R package *corMLPE* to evaluate the relationship between genetic differentiation (F_{ST}) and resistance distance (Clarke et al. 2002; Pinheiro and Bates 2012; Plue et al. 2019). We chose a GLS and MLPE approach to our linear mixed-effects models to account for heteroskedasticity in distance matrices and non-independent pairwise comparisons of populations, respectively. We created multiple alternative models with pairwise F_{ST} values (a measure of genetic differentiation and a proxy for gene flow; Gaggiotti et al. 1999) as the response variable, and four different predictor variables: 1) resistance distance of our landscape variables (IBR; independent variables and all cumulative variables), 2) presence/absence of river impoundments (IBB), 3) both resistance distance of landscape variables and presence/absence of river impoundments (IBR + IBB), and 4) river distance between sites (IBD). Resistance distance and river distance matrices were log-transformed to allow for comparison between models (Shirk et al. 2018; Razgour et al. 2021).

To assess the potential impacts of including geographically isolated populations in our analyses, which may inflate the influence of distance in our models or mask the influence of resistance among more connected populations, we examined data at two extents by including or excluding our most geographically isolated sites. Rockbridge Fork (site 2) and Hell For Certain Creek (site 7) are isolated geographically by distance and physically by a dam and reservoir, respectively, from all other examined populations. This isolation has contributed to the highest levels of genetic differentiation observed for examined populations of the KAD (Blanton et al. 2019). Our first extent included all sites, and the second extent included all sites except Rockbridge Fork (site 2) and Hell For Certain Creek (site 7). These two extents allowed us to identify landscape variables that contribute to the geographic and genetic isolation of sites separated by long distances with low expectations for connectivity as well as those restricting gene flow among populations with expected connectivity.

We calculated Akaike information criterion (AIC) values for all models to identify the top model that best describes the measured genetic differentiation among populations of KAD for each of our extents. To adjust for our limited number of samples (sites examined), we used sample size correction for the AIC values (AICc; Hurvich and Tsai 1989). We identified the top models as those with the lowest AICc score which differed from all other models by 2 or

more AICc values. To further evaluate model rankings, we considered AICc evidence weights and pseudo-R^2 using Nagelkerke's index to identify the best-supported models (Nagelkerke 1991; Razgour et al. 2021).

Results

Genetic diversity and differentiation

We collected 295 KAD fin clips from 15 sites, with 15–27 individuals per site (Table 1). All 295 individuals were successfully genotyped for the 11 microsatellite loci, with less than 0.03% missing data across all loci and individuals. No evidence of scoring errors due to null alleles, allele dropout, or stutter were found; therefore, all 11 loci were retained for analyses.

Three of 15 sites deviated from HWE ($p < 0.003$; Table 1), which may be expected since KAD have undergone population declines (Wigginton et al. 2005). A_R ranged from 1.4–6.2, with the lowest A_R in Rockbridge Fork (site 2; Table 1; Fig. 4) and Hell For Certain Creek (site 7) and highest in Middle Fork Quicksand Creek (site 5; Table S2). Rockbridge Fork (site 2; $H_o = 0.08$) and Hell For Certain Creek (site 7; $H_o = 0.26$) also had the lowest observed heterozygosity of all sites, with the highest H_o found in Wild Dog Creek (site 1) and Middle Fork Quicksand Creek (site 5), with $H_o \geq 0.65$ (Table 1). Significant inbreeding (F_{IS}) was identified in two sites, Hell For Certain Creek (site 7) and Big Double Creek (site 10), and outbreeding was found in one site, Laurel Fork (site 8; Table 1).

All pairwise F_{ST} values deviated significantly from panmixia ($p \leq 0.001$). Genetic differentiation was highest between Rockbridge Fork (site 2) and Hell For Certain Creek (site 7) ($F_{ST} = 0.68$; $p < 0.001$) and in comparisons of each of these two sites to all other sites examined ($F_{ST} \geq 0.29$; $p < 0.001$; Table 2). Pairwise comparisons of sites within the South Fork Kentucky River drainage showed some of the lowest, but still significant, levels of genetic differentiation ($F_{ST} < 0.15$; $p < 0.001$; Table 2).

Table 2 Pairwise F_{ST} values for the 15 populations examined for the Kentucky Arrow Darter (*Etheostoma spilotum*). All pairwise comparisons were significant after Bonferonni correction ($p \leq 0.003$)

Sites by Drainage	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Sturgeon Creek															
1. Wild Dog Creek	-														
Red River															
2. Rockbridge Fork	0.51	-													
North Fork Kentucky River															
3. Hell Creek	0.18	0.62	-												
4. Frozen Creek	0.24	0.56	0.26	-											
5. Middle Fork Quicksand Creek	0.10	0.46	0.12	0.12	-										
6. Clemons Fork	0.25	0.58	0.35	0.26	0.21	-									
Middle Fork Kentucky River															
7. Hell For Certain Creek	0.29	0.68	0.39	0.42	0.29	0.44	-								
South Fork Kentucky River															
8. Laurel Fork	0.16	0.54	0.23	0.26	0.15	0.31	0.40	-							
9. Bullskin Creek	0.22	0.65	0.22	0.36	0.17	0.39	0.47	0.22	-						
10. Big Double Creek	0.19	0.59	0.28	0.28	0.17	0.37	0.43	0.17	0.27	-					
11. Gilbert's Big Creek	0.20	0.60	0.29	0.29	0.18	0.38	0.42	0.19	0.34	0.10	-				
12. Elisha Creek	0.22	0.65	0.33	0.36	0.23	0.37	0.43	0.25	0.37	0.19	0.10	-			
13. Bowen Creek	0.22	0.60	0.33	0.28	0.21	0.38	0.44	0.23	0.35	0.15	0.10	0.16	-		
14. Spring Creek	0.25	0.63	0.37	0.36	0.25	0.40	0.46	0.26	0.38	0.18	0.14	0.13	0.16	-	
15. Upper Bear Creek	0.23	0.65	0.36	0.34	0.24	0.45	0.49	0.29	0.37	0.20	0.23	0.36	0.26	0.34	-

Isolation by Distance (IBD)

Our Mantel test demonstrated that river-km distance was significantly and positively correlated with F_{ST} between sites ($R^2 = 0.49$, $p < 0.01$; Fig. 3), supporting IBD as a contributing factor to population differentiation in KAD, as expected. Rockbridge Fork (site 2) and Hell For Certain Creek (site 7), two of our most geographically isolated populations also have some of the highest F_{ST} values compared to each other and to all other sites ($F_{ST} \geq 0.29$, $p < 0.001$; Table 2; Fig. 3). However, more than half the variation we see in F_{ST} values is unexplained by distance alone, suggesting genetic isolation in KAD is also influenced by other factors. For example, proximate sites in the South Fork Kentucky River system had higher than expected F_{ST} values given the short river-km distance separating them (Table 2; Fig. 3).

Riverscape models

All single variable and cumulative models tested at both extents, except for the model for dams alone, had high predictive power for the genetic differentiation (F_{ST}) observed in KAD (pseudo-$R^2 \geq 0.61$; Tables 3 and 4). For our extent including all sites, we recovered four models where ΔAICc was < 2 between models, indicating each of these four models are equally the best model given the data (Burnham and Anderson 2004; Table 3). Out of these four models, the model with the lowest AICc and highest AICc weight included percent forest cover and percent agriculture (AICc = -353.33, $\beta = 0.10$, SE = 0.007, and $p < 0.001$; Table 3), which indicates the cumulative effect of these two variables are primary drivers of higher genetic fixation (F_{ST}) among all extant KAD sites. However, since the top four models had a ΔAICc < 2, other land uses in addition to forest cover and agriculture likely also create resistance to dispersal and higher genetic differentiation between populations (Table 3).

By excluding Hell For Certain Creek (Site 7) and Rockbridge Fork (Site 2), we found percent forest cover best explained genetic differentiation between sites (AICc = -250.84, $\beta = 0.06$, SE = 0.005, and $p < 0.001$; Table 4). Percent forest cover had the highest w_t (0.45) and the ΔAICc was > 2 between this model and all others. Thus, at the extent

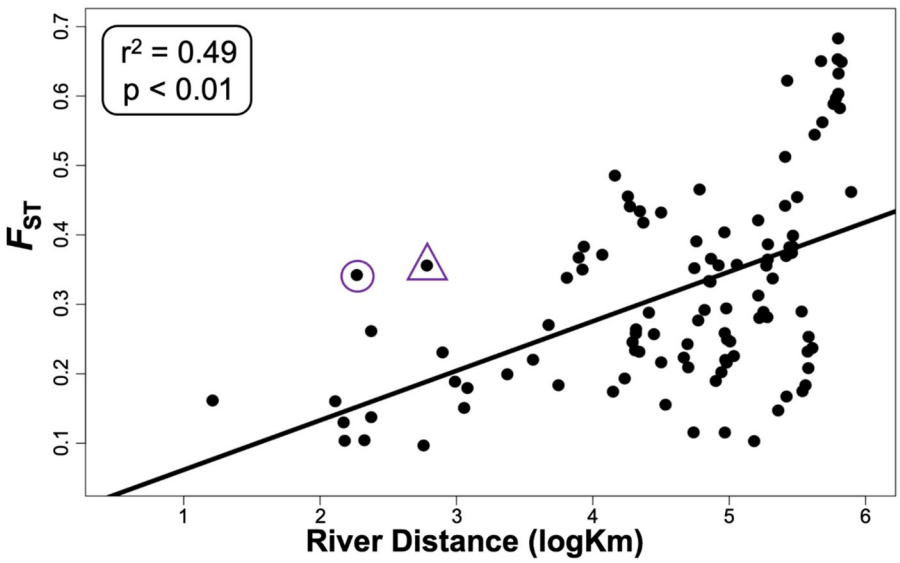

Fig. 3 Isolation by distance results displaying linear regression of genetic distance and river distance (black diagonal line) with corresponding results of a Mantel test ($r^2 = 0.49$, $p < 0.01$) for the 15 populations examined for the Kentucky Arrow Darter (*Etheostoma spilotum*) using F_{ST} genetic distances (Table 2) and log10 river kilometer pairwise distances. Points with a purple circle and purple triangle represent two sites in proximity that have a higher than expected F_{ST} given the short river kilometer distance between them. Purple circle represents Spring Creek (Site 14) and Upper Bear Creek (Site 15), purple triangle represents Elisha Creek (Site 12) and Upper Bear Creek (Site 15)

Table 3 Results of the maximum likelihood population effects (MLPE) riverscape genetics analysis for all 15 populations of the Kentucky Arrow Darter (*Etheostoma spilotum*) using F_{ST} as the measure of genetic differentiation. The table lists single variable and cumulative models tested by suite, model support based on Akaike information criterion corrected for small sample sizes (ΔAICc), log-likelihood (*LL*), number of model parameters (*k*), AICc weight (w_t), and pseudo-R^2. Suites are as follows: resistance distance of landscape variables (IBR), presence/absence of river impoundments (IBB), both resistance distance of landscape variables and presence/absence of river impoundments (IBR + IBB), isolation by distance (IBD). Model names are as follows: percent forest cover (Forest), percent agriculture area (Agriculture), percent urban area (Urban), percent surface mined area (Mining), river impoundment presence (Dam), Strahler stream order (Strahler), slope (Slope), and river distance (River Distance)

Suite	Model	ΔAICc	LL	k	w_t	Psuedo-R^2
IBR	Forest + Agriculture	0.00	180.87	5	0.29	0.71
	Agriculture + Urban + Forest	1.58	180.07	6	0.13	0.71
	Agriculture	3.59	179.07	4	0.05	0.70
	Forest	5.16	178.28	4	0.02	0.69
	Agriculture + Urban	5.22	178.26	5	0.02	0.69
	Agriculture + Forest + Strahler	8.96	176.38	6	0.00	0.68
	Agriculture + Urban + Forest + Strahler	9.16	176.28	7	0.00	0.68
	Agriculture + Urban + Mining + Forest	10.20	175.76	7	0.00	0.67
	Forest + Strahler	11.42	175.15	5	0.00	0.67
	Urban	13.54	174.10	4	0.00	0.66
	Agriculture + Mining	14.37	173.68	5	0.00	0.66
	Agriculture + Strahler	14.37	173.68	6	0.00	0.66
	Agriculture + Urban + Mining	15.22	173.25	6	0.00	0.65
	Strahler	19.05	171.34	4	0.00	0.64
	Slope + Strahler	19.63	171.05	5	0.00	0.63
	Slope	19.70	171.02	4	0.00	0.63
	Mining	20.56	170.58	4	0.00	0.63
IBB	Dam	108.12	126.81	4	0.00	0.02
IBR + IBB	Forest + Agriculture + Dam	0.16	181.89	6	0.27	0.72
	Agriculture + Urban + Forest + Dam	1.83	181.05	7	0.12	0.71
	Agriculture + Dam	3.48	180.23	5	0.05	0.71
	Agriculture + Urban + Dam	5.34	179.30	6	0.02	0.70
	Forest + Dam	6.29	178.82	5	0.01	0.70
	Agriculture + Forest + Strahler + Dam	9.04	177.45	7	0.00	0.69
	Agriculture + Urban + Forest + Strahler + Dam	9.25	177.34	8	0.00	0.69
	Agriculture + Urban + Mining + Forest + Dam	10.64	176.65	8	0.00	0.68
	Forest + Strahler + Dam	11.31	176.31	6	0.00	0.68
	Agriculture + Mining + Dam	13.61	175.16	6	0.00	0.67
	Agriculture + Urban + Mining + Forest + Dam	14.24	173.75	8	0.00	0.66
	Agriculture + Strahler + Dam	14.29	174.82	6	0.00	0.67
	Urban + Dam	14.51	174.71	5	0.00	0.67
	Agriculture + Urban + Mining + Dam	15.62	174.16	7	0.00	0.66
	Strahler + Dam	18.45	172.74	5	0.00	0.65
	Slope + Strahler + Dam	19.91	172.01	6	0.00	0.64
	Slope + Dam	20.10	171.92	5	0.00	0.64
	Mining + Dam	21.06	171.44	5	0.00	0.64
	Dam	108.12	126.81	4	0.00	0.02
IBD	River Distance	23.84	168.95	4	0.00	0.61

Table 4 Results of the maximum likelihood population effects (MLPE) riverscape genetics analysis for a subset of the Kentucky Arrow Darter (*Etheostoma spilotum*) populations that excludes Rockbridge Fork (Site 2) and Hell For Certain Creek (Site 7) using F_{ST} as the measure of genetic differentiation. The table lists single variable and cumulative models tested, model support based on Akaike information criterion corrected for small sample sizes (ΔAICc), log-likelihood (LL), model parameters AICc evidence weight (w_t), and pseudo-R^2. Suites are as follows: resistance distance of landscape variables (IBR), isolation by distance (IBD). Model names are as follows: percent forest cover (Forest), percent agriculture area (Agriculture), percent urban area (Urban), percent surface mined area (Mining), Strahler stream order (Strahler), slope (Slope), and river distance (River Distance)

Suite	Model	ΔAICc	LL	k	w_t	Psuedo-R^2
IBR	Forest	0.0	129.42	4	0.45	0.63
	Forest + Agriculture	2.1	128.37	5	0.16	0.62
	Agriculture + Forest + Urban	2.6	128.15	6	0.13	0.62
	Forest + Strahler	4.1	127.36	5	0.06	0.61
	Agriculture + Forest + Strahler	4.2	127.33	6	0.06	0.61
	Agriculture + Urban + Forest + Strahler	4.6	127.12	7	0.05	0.61
	Agriculture + Urban + Mining + Forest	5.9	126.47	7	0.02	0.60
	Agriculture	6.8	126.04	4	0.02	0.59
	Agriculture + Urban	6.9	125.96	5	0.01	0.59
	Agriculture + Strahler	7.5	125.68	5	0.01	0.59
	Slope + Strahler	7.9	125.46	5	0.01	0.59
	Agriculture + Mining	8.4	125.24	5	0.01	0.58
	Urban	8.8	125.05	4	0.01	0.58
	Slope	8.8	125.03	4	0.01	0.58
	Agriculture + Urban + Mining	9.2	124.84	6	0.00	0.58
	Strahler	9.2	124.80	4	0.00	0.58
	Mining	11.7	123.56	4	0.00	0.56
IBD	River Distance	8.6	125.13	4	0.01	0.58

excluding sites with low expectations for connectivity due to geographic isolation, percent forest cover best explains variance in gene flow among sites with high expectations for population connectivity (Table 4), where decreased percent forest cover contributes to increased genetic differentiation (F_{ST}).

River distance was not the top model and had a large ΔAICc when compared to top models at both extents (ΔAICc ≥ 8.6; Table 3; Table 4), supporting the hypothesis that KAD population connectivity, or lack thereof, is predominantly driven by IBR instead of the assumed IBD hypothesis. Although impoundments did not appear in the identified top model for our extent for all sites, there were two models which included impoundments that had ΔAICc < 2, suggesting that we could not reject the IBB hypothesis as an explanation for variance in gene flow across KAD sites range-wide (Table 3).

Discussion

Declines in matrix, habitat, or corridor quality contribute to population loss and reduce successful dispersal (and gene flow), lowering overall population connectivity (MacArthur & Wilson 1967; Hanski 1991; Leimar and Norberg 1997). Since reducing gene flow between populations can result in inbreeding and loss of genetic diversity, identifying landscape features and uses that reduce dispersal attempts or success is crucial to development of targeted conservation actions that improve population connectivity and persistence (Crooks & Sanjayan 2006; Castillo et al. 2016). We used a riverscape genetics approach to examine the effects of natural and anthropogenic landscape variables on population connectivity of KAD at sites with low expectations for connectivity and for populations with expected connectivity. Our

results indicated that genetic differentiation among populations of KAD is best explained, under both scenarios, by isolation by resistance (IBR). Specifically, we found that decreased forest cover best explains high genetic differentiation in extant KAD populations, with the cumulative effects of both decreased forest cover and anthropogenic land uses contributing to genetic isolation at sites with low expectations of connectivity due to geographic or physical (impoundment) isolation.

Impacts of forest cover on species occurrence and population connectivity

We provide the first evidence that human land uses, particularly those that reduce forest cover to less than 90% in a catchment, contribute to isolation by resistance and population differentiation in an imperiled, benthic and headwater-adapted fish. Over the past four centuries, the region where KAD occurs has experienced drastic loss of forest cover (Li et al. 2022; Figure S1). According to data from a land use reconstruction study, catchments in the distribution of KAD were likely completely forested before European settlement (Li et al. 2022; Figure S1). However, only a third of our catchments had over 90% forest cover, demonstrating that much of landscape where KAD occurs has experienced substantial forest cover loss. Our models identify declines in forest cover in a catchment as a key factor in gene flow reduction among populations of KAD, suggesting that most catchments do not have enough forest cover for the species to successfully disperse (Fig. 1). The negative influence of decreased forest cover on population connectivity in KAD aligns with previous occupancy studies that demonstrated 90% or greater forest cover in a given catchment was linked to increases in KAD occurrence and abundance (Hitt et al. 2016). Similarly, occurrences of other darters, including Kanawha Darter, *E. kanawhae*, and Holiday Darter, *E. brevirostrum*, are positively associated with the extent of forest cover in upstream catchments (Anderson et al. 2012; Pugh et al. 2021). For these species and KAD, lack of forest cover leads to habitat degradation by increasing sedimentation among other factors, which ultimately reduces species occurrences (Anderson et al. 2012; Hitt et al. 2016; Pugh et al. 2021).

Although our study was the first to show that loss of forest cover leads to increased genetic differentiation in a darter species, the relationship between forest cover and genetic differentiation has been demonstrated by landscape genetics studies in other imperiled or declining animal species. For example, Baden et al. (2019) demonstrated that deforestation from human activities causes high genetic structure and limits dispersal in a critically endangered Black-and-white Ruffed Lemur, *Varecia variegata*, by eliminating and degrading habitat and corridors. Similarly, Emel et al. (2019) found that the negative effects of habitat loss and fragmentation from timber harvest prevents successful dispersal of two narrowly distributed torrent salamander species, *Rhyacotriton kezeri* and *Rhyacotriton variegatus.*

Effects of anthropogenic land use on matrix condition

The association between forest cover and occurrence, abundance, and population genetic connectivity of small, headwater-adapted benthic fishes is most likely driven by changes to instream chemical and physical characteristics that result from forest removal or alteration. The river continuum concept suggests that the quality of the surrounding terrestrial matrix of a stream has the greatest impact in headwater streams, like those where KAD occur, where riparian forest cover plays a key role in the supply of organic material and ultimately controls stream inputs, productivity, and food resources (Vannote et al. 1980). In addition to supplying crucial ecosystem services from the riparian area around streams, forests play a key role at the catchment level. Even small losses of forest cover at the catchment level can increase runoff into streams, ultimately increasing sedimentation, degrading stream beds, and raising stream temperatures (Chang and Psaris 2013; Brito et al. 2020). KAD occurrence is negatively affected by increased stream temperature and sedimentation, suggesting the effects of decreased forest cover at both the riparian and catchment level are creating unfavorable conditions for the species (Hitt et al. 2016; Thomas et al. 2023). Ultimately, the effects of deforestation have degraded matrix, habitat, and corridor quality across KAD's range, creating high resistance for dispersal. Reduced dispersal potential in KAD prevents healthy metapopulation

dynamics, recovery of extirpated populations, and increases isolation of existing populations (Harrison 1991).

The loss of forest cover across KAD's range has been attributed to land use practices including agriculture, surface coal mining, and urbanization. Although each of these land use types negatively impact aquatic habitats, the nature of these impacts varies by land use type. For example, all contribute to increased sedimentation from removal of vegetative cover, but surface mining also contributes to elevated stream conductivity, agriculture to increased pesticides and herbicides, and urbanization to increased run-off and changes to normal stream flow dynamics, among others (Pond et al. 2008; Hitt et al. 2016, 2020). We did not find substantial support that our land use variables contributing to deforestation, including agriculture, surface coal mining, and urbanization, explained population isolation in KAD across the species distribution. However, this may reflect a limitation in our models, where our small sample sizes, due to the rarity of KAD, did not allow us to subset our models to account for the variation in predominant anthropogenic activities driving deforestation among tributary systems (Fig. 1). For example, forest loss in the North Fork Kentucky River and Middle Fork Kentucky River drainages is predominantly attributed to surface coal mining (Fig. 1). However, lack of forest cover in the northwestern areas of its range (e.g., Wild Dog Creek [Site 1] and Rockbridge Fork [Site 2]) is best explained by high levels of agriculture and urbanization (Fig. 1). The South Fork Kentucky River drainage, considered a stronghold for KAD based on population census size estimates and number of extant populations, has the most forest cover remaining across the range, but increased stream conductivity from surface coal mining in the area has been linked to decreases in gene flow among KAD populations (Hitt et al. 2016; Watson et al. 2024; Fig. 1). Therefore, given the spatial variation of land use types across the range of KAD, it is likely that the observed result of deforestation as the best indicator of high genetic differentiation in extant KAD populations reflects a combination of underlying habitat impacts stemming from different land use types, other than loss of forest cover alone.

Conservation implications and recommendations

Given the levels of population differentiation and population loss observed for KAD, we propose that deforestation from anthropogenic land uses has created unfavorable matrix, habitat, and corridor conditions for population persistence and successful dispersal sufficient to maintain population connectivity across most of KAD's range. We see evidence of increased population differentiation and isolation where deforestation in a watershed > 90%. Since only a third of the watersheds in our study area have over 90% forest cover, conditions are not favorable for successful dispersal and gene flow across much of the species' range (Fig. 1). Population isolation and declines in dispersal negatively impact population persistence and raise the risk of extinction (MacArthur and Wilson 1967; Hanski 1994); such species are susceptible to the extinction vortex, where small, isolated populations inbreed and lose genetic diversity, ultimately reducing species adaptability, survival, and reproductive success that leads to further decreases in population size and increased vulnerability to stochastic events (Fagan and Holmes 2006). Genetic metrics, including evidence of inbreeding, low diversity, and deviation from Hardy–Weinberg Equilibrium, indicate that KAD is already showing signs of the extinction vortex (Table 1). Therefore, reduced dispersal may increase risk of extinction by further decreasing population size, persistence, and reducing recolonization capability, especially as distance between extant KAD populations increases and number of populations decrease.

Increased local extinctions due to population isolation and decreases in patch occupancy have been well documented across many taxa, suggesting that successful dispersal encourages species persistence and avoidance of overall extinction through a mechanism known as the 'rescue effect' (Brown and Kodric-Brown 1977; Fahrig & Merriam 1985; Hanski 1986; Fitzpatrick et al. 2023). Therefore, conservation efforts for species experiencing increased geographic isolation among populations should focus on increasing number of patches and decreasing distance between patches, and restoring corridor health to promote species dispersal, thus encouraging patch recolonization and persistence. We recommend that conservation efforts for KAD focus on improving matrix and habitat conditions, specifically focusing

on mitigating current anthropogenic land use that contributes to further deforestation in watersheds where the species persists. For example, surface coal mining reclamation practices currently are considered ineffective at mitigating legacy effects of surface mining on the environment, with degradation of stream chemical properties persisting at least two decades after reclamation is completed (Lindberg et al. 2011; Hopkins et al. 2013). We recommend current reclamation processes be revised to better mitigate the legacy effects of coal surface mining on the environment and more importantly, future surface coal mining activities be restricted. Since the predominant anthropogenic land use driving deforestation varies throughout KAD's distribution (Fig. 1), it is crucial that mitigation and prevention is addressed at the region level. Therefore, conservation efforts should incorporate region-specific plans for land use mitigation across KAD's range, focusing on increasing forest cover by mitigating surface mining, agriculture, and urbanization.

Since the effects of mitigation and prevention of anthropogenic land use will most likely not occur in a time frame quick enough to prevent continued declines in number of populations and increased distance between populations, we suggest conservation management plans include genetic rescue, where captively propagated individuals are reintroduced to high quality or restored habitats, particularly those from which KAD populations have been extirpated (Ingvarsson 2001; Hedrick and Fredrickson 2010). Specifically, reintroduction efforts should choose localities that reduce distances between current populations, aiming to improve population connectivity and promote recolonization among other historic localities. To ensure long-term stability of extant populations, translocation of individuals among historically connected populations can assist in gene flow and ultimately promote increases in genetic diversity of the species (Fitzpatrick et al. 2023). The management framework for reintroduction efforts and assisted gene flow should follow guidelines set forth by George et al. (2009) to ensure success. Long-term genetic monitoring of KAD will support a management framework that can adapt to any genetic changes of the species, such as genetic swamping or outbreeding, instigated by conservation efforts, further anthropogenic land use activities, or stochastic events (George et al. 2009). By including long-term genetic monitoring with an adaptive management framework coupled with anthropogenic land use prevention and mitigation, conservation efforts will encourage long-term stability of the species.

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Data availability Data is available from the corresponding author upon request.

Declarations

Competing interests The authors have no relevant financial or non-financial interests to disclose.The authors declare no competing interests.

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References

Adamack AT, Gruber B (2014) PopGenReport: simplifying basic population genetic analyses in R. Methods Ecol Evol 5:384–387

Anderson GB, Freeman MC, Hagler MM, Freeman BJ (2012) Occupancy modeling and estimation of the holiday darter species complex within the etowah river system. Trans Am Fish Soc 141:34–45

Baden AL, Mancini AN, Federman S et al (2019) Anthropogenic pressures drive population genetic structuring across a Critically Endangered lemur species range. Sci Rep 9:16276

Bailey RM (1948) Status, relationships, and characters of the percid fish, *Poecilichthys sagitta* Jordan and Swain. Copeia 1948:77

Barletta M, Jaureguizar AJ, Baigun C et al (2010) Fish and aquatic habitat conservation in South America: a continental overview with emphasis on neotropical systems. J Fish Biol 76:2118–2176

Barnett ZC, Adams SB, Ochs CA, Garrick RC (2020) Crayfish populations genetically fragmented in streams impounded for 36–104 years. Freshw Biol 65:768–785

Baxter RM (1977) Environmental Effects of Dams and Impoundments. Annu Rev Ecol Evol Syst 8:255–283

Benz GW, Collins DE (1997) Aquatic fauna in peril: the southeastern perspective. Lenz Design & Communications, Georgia

Bland LM (2017) Global correlates of extinction risk in freshwater crayfish. Anim Conserv 20:532–542

Blanton RE, Cashner MF, Thomas MR et al (2019) Increased habitat fragmentation leads to isolation among and low genetic diversity within populations of the imperiled Kentucky Arrow Darter (*Etheostoma sagitta spilotum*). Conserv Genet 20:1009–1022

Bonada N, Múrria C, Zamora-Muñoz C et al (2009) Using community and population approaches to understand how contemporary and historical factors have shaped species distribution in river ecosystems. Glob Ecol Biogeogr 18:202–213

Brauer CJ, Beheregaray LB (2020) Recent and rapid anthropogenic habitat fragmentation increases extinction risk for freshwater biodiversity. Evol Appl 13:2857–2869

Brito JG, Roque FO, Martins RT et al (2020) Small forest losses degrade stream macroinvertebrate assemblages in the eastern Brazilian Amazon. Biol Conserv 241:108263

Brown JH, Kodric-Brown A (1977) Turnover rates in insular biogeography: effect of immigration on extinction. Ecology 58:445–449

Burnham KP, Anderson DR (2004) Multimodel inference: understanding AIC and BIC in model selection. Sociol Methods Res 33:261–304

Cabin RJ, Mitchell RJ (2000) To bonferroni or not to bonferroni: when and how are the questions. Bull Ecol Soc Am 81:246–248

Camak DT, Piller KR (2018) Going with the flow: testing the role of habitat isolation among three ecologically divergent darter species. Copeia 106:375–387

Castillo JA, Epps CW, Jeffress MR et al (2016) Replicated landscape genetic and network analyses reveal wide variation in functional connectivity for American pikas. Ecol Appl 26:1660–1676

Chang H, Psaris M (2013) Local landscape predictors of maximum stream temperature and thermal sensitivity in the Columbia River Basin, USA. Sci Total Environ 461–462:587–600

Cianciolo TR, McLaughlin DL, Zipper CE et al (2020) Impacts to water quality and biota persist in mining-influenced Appalachian streams. Sci Total Environ 717:137216

Clarke RT, Rothery P, Raybould AF (2002) Confidence limits for regression relationships between distance matrices: Estimating gene flow with distance. J Agric Biol Environ Stat 7:361–372

Coulon A, Fitzpatrick JW, Bowman R, Lovette IJ (2010) Effects of habitat fragmentation on effective dispersal of florida scrub-jays. Conserv Biol 24:1080–1088

Crooks KR, Sanjayan M (2006) Maintaining connections for nature. In: Crooks KR, Sanjayan M (eds) Connectivity Conservation, 1st edn. Cambridge University Press, pp 1–20

Dewitz J, U.S. Geological Survey (2024) National Land Cover Database (NLCD) 2019 Products

Dodd CK (1990) Effects of habitat fragmentation on a stream-dwelling species, the flattened musk turtle Sternotherus depressus. Biol Conserv 54:33–45

Dudgeon D, Arthington AH, Gessner MO et al (2006) Freshwater biodiversity: importance, threats, status and conservation challenges. Biol Rev 81:163–182

Ebenhard T (1991) Colonization in metapopulations: a review of theory and observations. Biol J Linn Soc 42:105–121

Emel SL, Olson DH, Knowles LL, Storfer A (2019) Comparative landscape genetics of two endemic torrent salamander species, *Rhyacotriton kezeri* and *R. variegatus*: implications for forest management and species conservation. Conserv Genet 20:801–815

England LE, Rosemond AD (2004) Small reductions in forest cover weaken terrestrial-aquatic linkages in headwater streams. Freshw Biol 49:721–734

ESRI 2011. ArcGIS Desktop: Release 10. Redlands, CA: Environmental Systems Research Institute.

Euclide P, Marsden JE (2018) Role of drainage and barriers in the genetic structuring of a tessellated darter population. Conserv Genet 19:1379–1392

Ewers RM, Didham RK (2007) Confounding factors in the detection of species responses to habitat fragmentation. Biol Rev 81:117–142

Fagan WF (2002) Connectivity, fragmentation, and extinction risk in dendritic metapopulations. Ecology 83:3243–3249

Fagan WF, Holmes EE (2006) Quantifying the extinction vortex. Ecol Lett 9:51–60

Fahrig L, Merriam G (1985) Habitat patch connectivity and population survival. Ecology 66:1762–1768

Farneda FZ, Rocha R, López-Baucells A et al (2015) Trait-related responses to habitat fragmentation in Amazonian bats. J Appl Ecol 52:1381–1391

Fausch KD, Torgersen CE, Baxter CV, Li HW (2002) Landscapes to riverscapes: bridging the gap between research and conservation of stream fishes: a continuous view of the river is needed to understand how processes interacting among scales set the context for stream fishes and their habitat. Bioscience 52:483

Fitzpatrick SW, Mittan-Moreau C, Miller M, Judson JM (2023) Genetic rescue remains underused for aiding recovery of federally listed vertebrates in the United States. J Hered 114:354–366

Fletcher RJ, Reichert BE, Holmes K (2018) The negative effects of habitat fragmentation operate at the scale of dispersal. Ecology 99:2176–2186

Floyd M (2022) Kentucky Arrow Darter Status Review: Summary and Evaluation. United States Fish and wildlife service southeast region, Kentucky ecological services field office, Frankfort, Kentucky

Fluker BL, Kuhajda BR, Harris PM (2014) The effects of riverine impoundment on genetic structure and gene flow in two stream fishes in the Mobile River basin. Freshw Biol 59:526–543

Frankham R (2005) Genetics and extinction. Biol Conserv 126:131–140

Fuller MR, Doyle MW, Strayer DL (2015) Causes and consequences of habitat fragmentation in river networks. Ann NY Acad Sci 1355:31–51

Gaggiotti OE, Lange O, Rassmann K, Gliddon C (1999) A comparison of two indirect methods for estimating average levels of gene flow using microsatellite data. Mol Ecol 8:1513–1520

Gallagher AJ, Hammerschlag N, Cooke SJ et al (2015) Evolutionary theory as a tool for predicting extinction risk. Trends Ecol Evol 30:61–65

George AL, Kuhajda BR, Williams JD et al (2009) Guidelines for propagation and translocation for freshwater fish conservation. Fisheries 34:529–545

Gilbert CH (1887) Descriptions of new and little known etheostomoids. Proc US Natl Mus 10:47–64

Goudet J (2005) HIERFSTAT, a package for r to compute and test hierarchical *F*-statistics. Mol Ecol Notes 5:184–186

Graf WL (1999) Dam nation: a geographic census of American dams and their large-scale hydrologic impacts. Water Resour Res 35:1305–1311

Greenwood MJ, Harding JS, Niyogi DK, McIntosh AR (2012) Improving the effectiveness of riparian management for aquatic invertebrates in a degraded agricultural landscape: stream size and land-use legacies. J Appl Ecol 49:213–222

Grummer JA, Beheregaray LB, Bernatchez L et al (2019) Aquatic landscape genomics and environmental effects on genetic variation. Trends Ecol Evol 34:641–654

Hanski I (1986) Population dynamics of shrews on small islands accord with the equilibrium model. Biol J Linn Soc 28:23–36

Hanski I (1991) Single-species metapopulation dynamics: concepts, models and observations. Biol J Linn Soc 42:17–38

Hanski I (1994) A Practical Model of Metapopulation Dynamics. J Anim Ecol 63:151

Harrison S (1991) Local extinction in a metapopulation context: an empirical evaluation. Biol J Linn Soc 42:73–88

Harrison S, Bruna E (1999) Habitat fragmentation and large-scale conservation: what do we know for sure? Ecography 22:225–232

Harrison S, Taylor AD (1997) Empirical evidence for metapopulation dynamics. In: Metapopulation biology. Elsevier, pp 27–42

Hedrick PW, Fredrickson R (2010) Genetic rescue guidelines with examples from Mexican wolves and Florida panthers. Conserv Genet 11:615–626

Hitt NP, Floyd M, Compton M, McDonald K (2016) Threshold Responses of Blackside Dace (*Chrosomus cumberlandensis*) and Kentucky Arrow Darter (*Etheostoma spilotum*) to Stream Conductivity. Southeast Nat 15:41–60

Hitt NP, Rogers KM, Kessler K, Macmillan HE (2020) Modeling occupancy of rare stream fish species in the upper Cumberland and Kentucky River Basins. Reston, VA

Hopkins RL, Altier BM, Haselman D et al (2013) Exploring the legacy effects of surface coal mining on stream chemistry. Hydrobiologia 713:87–95

Hurvich CM, Tsai C-L (1989) Regression and time series model selection in small samples. Biometrika 76:297–307

Ingvarsson PK (2001) Restoration of genetic variation lost – the genetic rescue hypothesis. Trends Ecol Evol 16:62–63

Jensen JL, Bohonak AJ, Kelley ST (2005) Isolation by distance, web service. BMC Genet 6:13. https://doi.org/10.1186/1471-2156-6-13

Jombart T (2008) *adegenet*: a R package for the multivariate analysis of genetic markers. Bioinformatics 24:1403–1405

Keenan K, McGinnity P, Cross TF et al (2013) diveRsity: An R package for the estimation and exploration of population genetics parameters and their associated errors. Methods Ecol Evol 4:782–788

Keinath DA, Doak DF, Hodges KE et al (2017) A global analysis of traits predicting species sensitivity to habitat fragmentation. Global Ecol Biogeogr 26:115–127

Keller LF, Waller DM (2002) Inbreeding effects in wild populations. Trends Ecol Evol 17:230–241

Kimura M, Weiss GH (1964) The Stepping Stone Model of Population Structure and the Decrease of Genetic Correlation With Distance. Genetics 49:561–576

Kreye M, Adams D, Escobedo F (2014) The value of forest conservation for water quality protection. Forests 5:862–884

Li X, Tian H, Pan S, Lu C (2022) Land use and land cover changes in the contiguous United States from 1630 to 2020. https://doi.org/10.5281/zenodo.7055086

Lindberg TT, Bernhardt ES, Bier R et al (2011) Cumulative impacts of mountaintop mining on an Appalachian watershed. PNAS 108:20929–20934

MacArthur RH, Wilson EO (1967) The Theory of Island Biogeography. Princeton University Press, Princeton, NJ

Magalhães MF, Batalha DC, Collares-Pereira MJ (2002) Gradients in stream fish assemblages across a Mediterranean landscape: contributions of environmental factors and spatial structure. Freshw Biol 47:1015–1031

Manel S, Schwartz MK, Luikart G, Taberlet P (2003) Landscape genetics: combining landscape ecology and population genetics. Trends Ecol Evol 18:189–197

Mbaka JG, Wanjiru Mwaniki M (2015) A global review of the downstream effects of small impoundments on stream habitat conditions and macroinvertebrates. Environ Rev 23:257–262

McCullough DR, Society Wildlife (eds) (1996) Metapopulations and wildlife conservation. Island Press, Washington, D.C

McRae BH (2006) Isolation by Resistance. Evolution 60:1551–1561

Méndez M, Vögeli M, Tella JL, Godoy JA (2014) Joint effects of population size and isolation on genetic erosion in fragmented populations: finding fragmentation thresholds for management. Evol Appl 7:506–518

Montoya-Burgos JI (2003) Historical biogeography of the catfish genus *Hypostomus* (Siluriformes: Loricariidae), with implications on the diversification of Neotropical ichthyofauna. Mol Ecol 12:1855–1867

Nabutanyi P, Wittmann MJ (2021) Models for eco-evolutionary extinction vortices under balancing selection. Am Nat 197:336–350

Nagelkerke NJD (1991) A note on a general definition of the coefficient of determination. Biometrika 78:691–692

O'Driscoll M, Clinton S, Jefferson A et al (2010) Urbanization effects on watershed hydrology and in-stream processes in the Southern United States. Water 2:605–648

Leimar O, Norberg U (1997) Metapopulation Extinction and Genetic Variation in Dispersal-Related Traits. Oikos 80:448

Page LM, Burr BM (2011) Peterson field guide to freshwater fishes of North America north of Mexico. Mariner Books, an imprint of HarperCollins Publishers, Boston

Pericak AA, Thomas CJ, Kroodsma DA et al (2018) Mapping the yearly extent of surface coal mining in Central Appalachia using Landsat and Google Earth Engine. PLoS ONE 13:e0197758

Pinheiro JC, Bates DJ (2012) Linear and Nonlinear Mixed Effects Models. R Package.

Plue J, Aavik T, Cousins SAO (2019) Grazing networks promote plant functional connectivity among isolated grassland communities. Divers Distrib 25:102–115

Pond GJ, Passmore ME, Borsuk FA et al (2008) Downstream effects of mountaintop coal mining: comparing biological conditions using family- and genus-level macroinvertebrate bioassessment tools. J N Am Benthol Soc 27:717–737

Pugh MW, Pandolfi G, Franklin T, Gangloff MM (2021) Influences of in-stream habitat and upstream land-use on site occupancy of the Kanawha darter (*Etheostoma kanawhae*): a narrowly distributed species from the New River (Upper Kanawha Basin). Aquat Conserv Mar Freshw Ecosyst 31:494–503

R Development Core Team (2022) R: a language and environment for statistical computing. Vienna, Austria. http://www.R-project.org/

Razgour O, Kasso M, Santos H, Juste J (2021) Up in the air: Threats to Afromontane biodiversity from climate change and habitat loss revealed by genetic monitoring of the Ethiopian Highlands bat. Evol Appl 14:794–806

Reis RE, Albert JS, Di Dario F et al (2016) Fish biodiversity and conservation in South America. J Fish Biol 89:12–47

Schippers P, Verboom J, Knaapen JP, Van Apeldoorn RC (1996) Dispersal and habitat connectivity in complex heterogeneous landscapes: an analysis with a GIS-based random walk model. Ecography 19:97–106

Shirk AJ, Landguth EL, Cushman SA (2018) A comparison of regression methods for model selection in individual-based landscape genetic analysis. Mol Ecol Resour 18:55–67

Slatkin M (1993) Isolation by distance in equilibrium and non-equilibrium populations. Evolution 47:264–279

Sotola VA, Schrey AW, Ragsdale AK et al (2017) Genetic evidence of isolation by distance and impact of impoundments on genetic diversity of riverine channel catfish. Trans Am Fish Soc 146:1204–1211

Stacey PB, Taper M (1992) environmental variation and the persistence of small populations. Ecol Appl 2:18–29

Stokes MF, Perron JT (2020) Modeling the evolution of aquatic organisms in dynamic river basins. J Geophys Res Earth Sur 125:e2020JF005652

Sutherland AB, Meyer JL, Gardiner EP (2002) Effects of land cover on sediment regime and fish assemblage structure in four southern Appalachian streams. Freshw Biol 47:1791–1805

Thomas MR, Blanton RE, Ghezelayagh A, Near TJ (2023) Species-level recognition and redescription of the kentucky arrow darter, etheostoma spilotum gilbert (Percidae: litocara). Bull Peabody Mus Nat Hist. https://doi.org/10.3374/014.064.0103

Thomas, MR (2008). Distribution and status of the Kentucky Arrow Darter (*Etheostoma sagitta spilotum*). Project Report, Kentucky Department of Fish and Wildlife Resources, Frankfort, KY, State and Tribal Wildlife Grant T-9-RSI-1.

Tyers, M (2017) River Network Distance Computation and Applications. R Package.

U.S. Geological Survey (2019) National Hydrography Dataset Plus High Resolution (NHDPlus HR). USGS National Map Downloadable Data Collection.

[USFWS] United States Fish and Wildlife Service (2016) Endangered and threatened wildlife and plants; threatened species status for Kentucky Arrow Darter with 4(d) Rule. Final Rule Federal Register 81(193):68963–68985

Van Etten J (2017) R Package gdistance: Distances and Routes on Geographical Grids. J Stat Soft 76:1–21

Vandermeer J, Carvajal R (2001) Metapopulation Dynamics and the Quality of the Matrix. Am Nat 158:211–220

Vannote RL, Minshall GW, Cummins KW et al (1980) The river continuum concept. Can J Fish Aquat Sci 37:130–137

Vrijenhoek RC (1998) Conservation genetics of freshwater fish. J Fish Biol 53:394–412

Wallus R, Simon TP (2006) Reproductive biology and early life history of fishes in the Ohio River drainage. CRC Press, Boca Raton

Warren ML, Burr BM, Walsh SJ et al (2000) Diversity, distribution, and conservation status of the native freshwater fishes of the southern United States. Fisheries 25:7–31

Watson RA, Culley AV, Haase CG et al (2024) Instream barriers contribute to population isolation of a small-bodied, benthic, headwater-specialist fish (Percidae). Ecol Freshw Fish 33:e12769

Weir BS, Cockerham CC (1984) Estimating F-statistics for the analysis of population structure. Evolution 38:1358
Wiens JJ, Donoghue MJ (2004) Historical biogeography, ecology and species richness. Trends Ecol Evol 19:639–644
Wigginton JE, Cutler DJ, Abecasis GR (2005) A note on exact tests of hardy-weinberg equilibrium. Am J Hum Genet 76:887–893
Winton RS, Calamita E, Wehrli B (2019) Reviews and syntheses: dams, water quality and tropical reservoir stratification. Biogeosciences 16:1657–1671
Wu J, Huang J, Han X et al (2003) Three-Gorges Dam-Experiment in Habitat Fragmentation? Science 300:1239–1240

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